Marcell Howard

PERSONAL DETAILS

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RESEARCH INTERESTS

Gravitational Particle Production, Early Universe Cosmology, Cosmological Gravitational Wave Backgrounds, Quantum Cosmology, Primordial Black Holes

ACADEMIC APPOINTMENTS

Postdoctoral Research Associate *2026 - Present*
Brown University
Advised by Stephon Alexander and Cyril Creque-Sarbinowski

EDUCATION

University of Pittsburgh, Pittsburgh, PA *2019 - 2026*
Ph.D., Physics

Case Western Reserve University, Cleveland, OH *2015 - 2019*
B.S., Mathematics and Physics

RESEARCH EXPERIENCE

Graduate Research Assistant *2020 - 2026*
University of Pittsburgh advised by Arthur Kosowsky
Doctoral Thesis: Issues of Gravitation in Cosmology

Senior Project *2018 - 2019*
Case Western Reserve University advised by Kurt Hinterbichler
Senior Project Thesis: The People vs Mimetic Gravity

Research Experience for Undergraduates *Summer 2018*
Rutgers, the State University of New Jersey advised by Andrew Baker
Probing the Evolution of Galaxies Using Stellar Mass Selected Samples

Research Experience for Undergraduates *Summer 2017*
Cornell University advised by Gordon Stacey
Using Far-Infrared Fine-Structure Lines for Characterizing Star Formation Processes in Nearby Galaxies

FELLOWSHIPS, SCHOLARSHIPS AND AWARDS

PITT PACC Fellowship *Summer 2026*
Competitive internal department fellowship for excellence in research and scholarship

dB-SERC Mentor-Mentee Evidence-Based Teaching Award *Fall 2025*
Competitive teaching award for evidence-based teaching with a focus on transforming at least a two-week segment of a natural science course to improve student learning outcomes in a measurable way

Provost's Dissertation Completion Fellowship - Fall 2025 *Declined*
Competitive university-wide fellowship that acknowledges academic and scholarly accomplishments and ensure dedicated time to focus full-time on finishing the dissertation

Peter F.M. Koehler Pre-Doctoral Fellowship *2023 - 2024*
Competitive internal department fellowship for excellence in teaching and research

2021 Honorable Mention, Ford Foundation Predoctoral Fellowship *2021*

Dietrich School of Arts and Sciences Summer Research Predoctoral Fellowship *2020*
Fellowship to support the most well-prepared students and speed the transition into the research that will become thesis work and to help meet diversity goals

Kenneth P. Dietrich School of Arts & Sciences Fellowship *2019*
For incoming graduate students for recognition of an outstanding undergraduate record

Case Alumni Junior/Senior Scholarship *2017 - 2019*
For rising sophomores and juniors pursuing a B.S. in engineering, math, or the applied sciences based on merit, need and personal skills

PUBLICATIONS LIST

* = *Corresponding Author*

M. Howard* and M. König, *Elastic scattering of cosmological gravitational wave backgrounds: primordial black holes and stellar objects*, J. Cosmology Astropart. Phys. **2024** (2024) 045 [2309.15925]

T. Daniel, M. Howard* and M. König, *An SZ-like effect on cosmological gravitational wave backgrounds*, J. Cosmology Astropart. Phys. **2023** (2023) 041 [2308.00111]

S. Alexander, T. Daniel, M. Howard* and M. König, *Exact fermionic Chern-Simons-Kodama state in quantum gravity*, Phys. Rev. D **106** (2022) 106012 [2207.11856]

M. Howard*, A. Kosowsky and G. Valogiannis, *Galaxy Cluster Statistics in Modified Gravity Cosmologies*, *arXiv e-prints* (2022) arXiv:2205.13015 [2205.13015]

SEMINARS AND COLLOQUIA

Brown University, STEMJazz *Nov 2026*

University of Maryland, EPT Seminar *Nov 2025*

Indiana University Bloomington, HEP Seminar *Nov 2025*

Rice University, Astronomy Undergraduate Seminar *Oct 2025*

University of Florida, HEP Seminar *Oct 2025*

University of Pittsburgh, Mathematics Graduate Student Seminar *Sep 2025*

University of Illinois Urbana-Champaign, HEP Seminar *Sep 2025*

University of Pittsburgh, AstroLunch *Jan 2025*

Pomona College Colloquium *Sep 2024*

Simons-NSBP Scholars Program, Flatiron Institute *July 2024*

Syracuse University Colloquium *Nov 2023*

Syracuse University, Watson Cosmology Group Seminar *Nov 2023*

MIT, String Theory Group Meeting *Feb 2023*

Case Western Reserve University, Particle Astrophysics Seminar *Sep 2022*

Brown University, Alexander Lab Group Meeting *Mar 2022*

SELECTED WORKSHOPS AND CONFERENCES

† = *Invited Talk*

33rd Texas Symposium on Relativistic Astrophysics *Dec 2025*

28th International Conference on Particle Physics and Cosmology *Oct 2025*

Division of Particles & Fields - Phenomenology Symposium 2024 *May 2024*

†**International Symposium on Particles, Strings, and Cosmology** *June 2023*

†**Ending Inflation and the Hot Big Bang Workshop**
Simons Center for Geometry and Physics *June 2023*

Pennsylvania State University, Neighborhood Workshop *Apr 2023*

American Physical Society April Meeting 2022 *Apr 2022*

National Society of Black Physicists Conference *Nov 2021*

TEACHING EXPERIENCE

Recitation Leader
PHYS2320: Quantum Theory of Fields II

Course Instructor
PHYS 0110: Introduction to Physics 1

Guest Lecturer
PHYS 3726: General Relativity 2
General Relativity as a Quantum Field Theory

Graduate Teaching Assistant
PHYS 0111: Introduction to Physics 2
PHYS 0174: Basic Physics for Science and Engineering 1
PHYS 0175: Basic Physics for Science and Engineering 2
PHYS 0212: Introduction to Laboratory Physics
PHYS 0476: Basic Physics for Science and Engineering II - Honors

- Held two-four recitation sessions per week.
- Graded quizzes and tests.

Graduate Core Course Departmental Tutor
PHYS 2565: Non-Relativistic Quantum Mechanics 1
PHYS 2566: Non-Relativistic Quantum Mechanics 2
PHYS 1373 & 2373 (Combined section): Mathematical Methods in Physics

- Held two office hours per week.
- Provided homework help as well as furthered conceptual understanding of the material.

Undergraduate Supplemental Instructor
MATH 121: Calculus for Scientists and Engineers I
MATH 122: Calculus for Scientists and Engineers II

- Held three small groups and one recitation session per week.
- Graded quizzes and tests.

Undergraduate Teaching Assistant
PHYS 121: General Physics I - Mechanics
PHYS 122: General Physics II - Electricity and Magnetism

- Graded written homework handed in every week.
- Handed out in-class assignments and provided help to various students.

OUTREACH AND SERVICE

Allegheny Observatory Public Lecture Series *July 2025*
Allegheny Observatory
Invited Presenter: The First Three Minutes

Exploring the Universe with Pitt Astronomers *Aug 2024*
University of Pittsburgh
Volunteer Presenter

Astronomy on Tap *Aug 2024*
Two Fray's Brewery
Volunteer Presenter: When Black Holes Aren't So Black

Astrophysicist for a Day *Mar 2024*
Allegheny Observatory
Volunteer Workshop Leader: Rediscovering Our Accelerating Universe

Society of Physics Students Graduate Student Panel *Dec 2023*
University of Pittsburgh
Panelist

Python Boot Camp 2023: Functions and Modules *May 2023*
University of Pittsburgh
Volunteer Presenter

Astronomy on Tap NYC *May 2023*
Pete's Candy Store
Invited Presenter: The Wave function of the Universe

International Baccalaureate Class Group Project *Mar 2023*
St Edward High School
Invited Speaker

Seminar at High School (SAHS) and Pathway Career Speaker Series *Dec 2022*
Woodland Hills High School
Invited Speaker

Society of Physics Students Graduate Student Panel *Oct 2022*
University of Pittsburgh
Panelist

Astronomy on Tap *Sept 2022*
Two Fray's Brewery
Volunteer Presenter: The Wave function of the Universe

Summer Seminar Series: So You Wanna Get Good At LaTeX *July 2022*
University of Pittsburgh
Volunteer Presenter

Cornell's Focus for Teens *Summer 2017*
Cornell University
Workshop Leader

MEDIA CONTRIBUTIONS AND APPEARANCES

Black In Cosmology *June 2025*
Nature Astronomy
https://doi.org/10.1038/s41550-025-02577-9

SKILLS

Programming Languages	Python, Mathematica, Matlab
Software	SciPy/matplotlib, AstroPy, git

PROFESSIONAL ORGANIZATIONS

Pittsburgh Particle Astrophysics and Cosmology Center *2020 - 2026*

National Society of Black Physicists *2018 - Present*

REFERENCES

Arthur Kosowsky
kosowsky[at]pitt.edu
University of Pittsburgh
Andrew Zentner
zentner[at]pitt.edu
University of Pittsburgh
Stephon Alexander
stephon_alexander[at]brown.edu
Brown University